# Supplementary Material: Trajectory Recovery via Constrained LLM Reasoning and Difficulty-Aware Routing

## A EXTENDED MULTI-SPLIT BENCHMARKS

Section 4.2 (Table 1) of the main paper reports end-to-end recovery performance across the five standard held-out splits on UrbanVehicle (UrbanVehicle Test $N = 400$, UrbanVehicle Val $N = 400$, and UrbanVehicle Large-Scale $N = 1{,}000$) and Carla (Carla Test $N = 185$ and Carla Val $N = 182$). This appendix details our cross-method evaluation protocol, reports the complete end-to-end multi-split benchmark ledger with absolute Macro F1 gains and relative 3D Vol-MSE reductions (Figure 1 and Table 1), and proves the timestamp-projection invariance property of LCSS under uncalibrated temporal arrival drift.

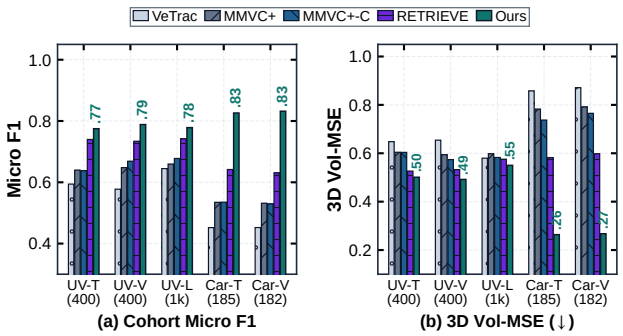



**Figure 1: Multi-split benchmark comparison complementing Figure 5a of the main paper: (a) Cohort-Merged Micro F1 and (b) 3D Spatio-Temporal Volume MSE (Vol-MSE) across all five standard splits on UrbanVehicle and Carla.**

### A.1 Cross-Method Evaluation Protocol

To ensure an aligned comparison across VeTrac, MMVC+, MMVC+-C, RETRIEVE, and our C1–C3 recovery framework (**Ours**), our unified evaluation pipeline enforces three strict protocol invariants:

(1) **Identical ground-truth vehicle cohorts.** Across every held-out split on UrbanVehicle (Test $N = 400$, Val $N = 400$, Large-Scale $N = 1{,}000$, and complete held-out triggered cohorts Test48 $N = 48$ and Val37 $N = 37$) and Carla (Test $N = 185$ and Val $N = 182$), all methods are evaluated against the *exact same ground-truth trajectory subset* $\mathcal{T}^*_{\text{sub}}$ with strict vehicle-source isolation from historical logs. For methods that decouple multi-vehicle visual clusters into sub-trajectories (RETRIEVE's refined 3D voxel sub-clusters and C2's decoupled corridor tracks on Carla), each reconstructed track is matched to its dominant ground-truth vehicle under the same $q(g)$ rule (main paper Section 2.3) so multi-vehicle clusters are scored without cross-ID truncation.

(2) **Unified five-metric accuracy and efficiency suite.** Every method is scored via identical metric definitions (Section 2.3 of the main paper): (i) per-vehicle Macro F1; (ii) cohort-merged Micro F1; (iii) bidirectional Spatio-Temporal Linear Combine (STLC, $\lambda = 0.5$); (iv) normalized Longest Common Subsequence (LCSS); and (v) 3D Spatio-Temporal Volume MSE (Vol-MSE over a $128^3$ spatiotemporal voxel grid), alongside per-query latency, token volume, API cost, and LLM invocation rate.

(3) **Complete camera-to-road projection and temporal alignment on Carla.** On Carla (20 fps simulation clock across 1,169 cameras: 426 intersection and 743 mid-block cameras on a 426-node road network), all methods operate on second-aligned timestamps ($t \in [0, 86{,}400]$ s) and the complete 1,169-camera Haversine road-node projection table, evaluated over the exact same $N = 182$ validation and $N = 185$ test ground-truth vehicle trajectories without cross-ID truncation. Under dense plate-free multi-camera sensing, unconditional shortest-path chaining (*SigCluster + Unconstrained Dijkstra*, 0.1581 / 0.1460 Macro F1 in Table 1 of the main paper) links co-clustered visual outliers and opposite-lane mid-block projections into long multi-hop detours across intersections, whereas C2 decouples intertwined tracks and enforces kinematic/zero-detour bounds before path expansion (**0.8139** / **0.8193** Macro F1).

### A.2 Timestamp-Projection Invariance of LCSS

Given a ground-truth trajectory $T^*_g = \{(u^*_{g,j}, [t^{\text{start}}_{g,j}, t^{\text{end}}_{g,j}])\}_{j=1}^{|T^*_g|}$ and its paired predicted trajectory $\hat{T}_q = \{(u_l, \hat{t}_l)\}_{l=1}^{|\hat{T}_q|}$ for $q = q(g)$:

(1) **Time-windowed node matching (Macro F1 and Micro F1):** A predicted point $(u_l, \hat{t}_l)$ counts as a hit $H_g$ if and only if $u_l = u^*_{g,j}$ and $t^{\text{start}}_{g,j} \le \hat{t}_l \le t^{\text{end}}_{g,j}$. Per-vehicle precision $P_g = H_g/|\hat{T}_q|$ (0 when $|\hat{T}_q| = 0$) and recall $R_g = H_g/|T^*_g|$ (as well as 3D Spatio-Temporal Volume MSE) penalize both false detour nodes and any interpolated node whose timestamp $\hat{t}_l$ misses the true arrival window $[t^{\text{start}}_{g,j}, t^{\text{end}}_{g,j}]$.

(2) **Timestamp-projection invariance of LCSS under temporal drift:** In the standard evaluation pipeline, LCSS is defined as the length of the longest common node subsequence normalized by $\max(|\hat{T}_q|, |T^*_g|)$:

$$\text{LCSS}(\hat{T}_q, T^*_g) = \frac{\left|\text{LCS}\left(\{u_l\}_{l=1}^{|\hat{T}_q|},\ \{u^*_{g,j}\}_{j=1}^{|T^*_g|}\right)\right|}{\max\left(|\hat{T}_q|,\ |T^*_g|\right)}. \tag{1}$$

Because $\text{LCSS}(\hat{T}_q, T^*_g)$ projects out all arrival timestamps $\hat{t}_l$, any trajectory $\hat{T}'_q = \{(u_l, \hat{t}_l + \Delta t_l)\}$ that traverses the corridor nodes in topological order under uncalibrated signal-delay or travel-time interpolation achieves $\text{LCSS}(\hat{T}'_q, T^*_g) = 1.0$ (for example, on UrbanVehicle Test vehicle `vid=35613`, MMVC+-C matches all 10/10 road nodes in sequence for LCSS = 1.000, yet 9 of 10 interpolated timestamps miss $[t^{\text{start}}_{g,j}, t^{\text{end}}_{g,j}]$, yielding $H_g = 1$ and $\text{F1}_g = 0.100$). Consequently, while **Ours** surpasses MMVC+-C on LCSS on UrbanVehicle Test and both Carla splits, MMVC+-C records a marginally higher untimed LCSS (+0.0017–0.0020)

**Table 1: Complete end-to-end comparison across the five standard held-out splits on UrbanVehicle (Test $N = 400$, Val $N = 400$, Large-Scale $N = 1{,}000$) and Carla (Test $N = 185$ and Val $N = 182$) under the unified five-metric suite, reporting absolute Macro F1 gains (ΔMacro F1) and relative 3D Vol-MSE reductions of Ours (C1–C3) over each baseline.**

| Dataset & Split ($N$) | Method | Macro F1 ↑ | Micro F1 ↑ | STLC ↑ | LCSS ↑ | Vol-MSE ↓ | Gain ΔMacro F1 | Rel. ΔVol-MSE |
|---|---|---|---|---|---|---|---|---|
| UrbanVehicle Test ($N = 400$) | VeTrac | 0.4501 | 0.5942 | 0.6902 | 0.4512 | 0.6479 | +0.3014 | −22.7% |
| | MMVC+ | 0.6025 | 0.6400 | 0.8495 | 0.7207 | 0.6042 | +0.1490 | −17.1% |
| | MMVC+-C | 0.6017 | 0.6376 | 0.8413 | 0.7308 | 0.6037 | +0.1498 | −17.0% |
| | RETRIEVE | 0.7164 | 0.7398 | 0.8552 | 0.6814 | 0.5263 | +0.0351 | −4.8% |
| | **Ours (C1–C3)** | **0.7515** | **0.7749** | **0.8793** | **0.7490** | **0.5010** | **Best** | **Best** |
| UrbanVehicle Val ($N = 400$) | VeTrac | 0.4373 | 0.5775 | 0.6807 | 0.4453 | 0.6542 | +0.3313 | −24.8% |
| | MMVC+ | 0.5991 | 0.6478 | 0.8482 | 0.7220 | 0.5939 | +0.1695 | −17.2% |
| | MMVC+-C | 0.6347 | 0.6688 | 0.8654 | **0.7660** | 0.5738 | +0.1339 | −14.3% |
| | RETRIEVE | 0.7136 | 0.7340 | 0.8550 | 0.6812 | 0.5322 | +0.0550 | −7.6% |
| | **Ours (C1–C3)** | **0.7686** | **0.7884** | **0.8908** | 0.7643 | **0.4919** | **Best** | **Best** |
| UrbanVehicle Large-Scale ($N = 1{,}000$) | VeTrac | 0.5532 | 0.6447 | 0.7656 | 0.5558 | 0.5797 | +0.1969 | −5.1% |
| | MMVC+ | 0.6159 | 0.6596 | 0.8405 | 0.6937 | 0.5979 | +0.1342 | −8.0% |
| | MMVC+-C | 0.6469 | 0.6778 | 0.8623 | **0.7380** | 0.5827 | +0.1032 | −5.6% |
| | RETRIEVE | 0.7178 | 0.7420 | 0.8522 | 0.6949 | 0.5754 | +0.0323 | −4.4% |
| | **Ours (C1–C3)** | **0.7501** | **0.7785** | **0.8751** | 0.7360 | **0.5501** | **Best** | **Best** |
| Carla Test ($N = 185$) | VeTrac | 0.4293 | 0.4521 | 0.7483 | 0.4976 | 0.8578 | +0.3846 | −69.3% |
| | MMVC+ | 0.5152 | 0.5347 | 0.7854 | 0.6056 | 0.7828 | +0.2987 | −66.3% |
| | MMVC+-C | 0.5183 | 0.5352 | 0.7576 | 0.6232 | 0.7373 | +0.2956 | −64.2% |
| | RETRIEVE | 0.6304 | 0.6417 | 0.8102 | 0.6310 | 0.5822 | +0.1835 | −54.7% |
| | **Ours (C1–C3)** | **0.8139** | **0.8262** | **0.8919** | **0.7643** | **0.2636** | **Best** | **Best** |
| Carla Val ($N = 182$) | VeTrac | 0.4239 | 0.4523 | 0.7428 | 0.5003 | 0.8712 | +0.3954 | −69.3% |
| | MMVC+ | 0.5072 | 0.5318 | 0.7773 | 0.6149 | 0.7914 | +0.3121 | −66.2% |
| | MMVC+-C | 0.5028 | 0.5294 | 0.7601 | 0.6135 | 0.7651 | +0.3165 | −65.0% |
| | RETRIEVE | 0.6110 | 0.6307 | 0.8101 | 0.6179 | 0.5976 | +0.2083 | −55.3% |
| | **Ours (C1–C3)** | **0.8193** | **0.8320** | **0.8972** | **0.7639** | **0.2674** | **Best** | **Best** |

on UrbanVehicle Val and Large-Scale while trailing **Ours** by 10.3–13.4 percentage points in Macro F1 and 5.6%–14.3% in 3D Vol-MSE on those two splits (Table 1).

# B THEORETICAL PROOFS

## B.1 Proof of Proposition 1

**Proof.** Let $O_q = (o_1, \ldots, o_m)$ be a triggered candidate cluster ($\gamma_{\text{inv}}(O_q) = 1$) with candidate local sub-interval bindings $\mathcal{B}_{\text{local}}$ retrieved from the source-disjoint bank $\mathcal{E}_{\text{bank}}$ ($s_e \neq s_q$). Consider any binding $b \in \mathcal{B}_{\text{local}}$ overriding camera $c_{i_b}$ from its Top-1 prior $v_{\text{pub}} = v_{c_{i_b},1}$ to secondary approach $v_{\text{exp}} = v_{c_{i_b},2}$:

(1) *Bidirectional clamping at interior observations ($bnd(b) = 0$):* When $1 < i_b < m$, node $v^*_{i_b}$ is simultaneously constrained by the upstream interval $P_{\text{SP}}(v^*_{i_b-1}, v^*_{i_b})$ and the downstream interval $P_{\text{SP}}(v^*_{i_b}, v^*_{i_b+1})$. Under C2's bounded Dijkstra execution, replacing $v_{\text{pub}}$ with $v_{\text{exp}}$ must preserve directed reachability and kinematic feasibility ($\hat{v} \le v_{\max}$) on both adjacent intervals, preventing unconstrained endpoint loss.

(2) *Open-boundary single-source truncation:* When $\text{bnd}(b) = 1$ and $\text{supp}(b) = 1$ at $i_b = 1$ (or $i_b = m$), the outer side of $c_{i_b}$ has no clamping anchor in $O_q$. If a single-source local match ($k = 1$) maps $c_1$ to a downstream approach $v_{\text{exp}}$ satisfying $P_{\text{SP}}(v_{\text{exp}}, v^*_2) \subsetneq P_{\text{SP}}(v_{\text{pub}}, v^*_2)$ (or maps $c_m$ to an upstream approach $v_{\text{exp}}$ with $d_{\text{SP}}(v^*_{m-1}, v_{\text{exp}}) < d_{\text{SP}}(v^*_{m-1}, v_{\text{pub}})$), C2 drops all links in $P_{\text{SP}}(v_{\text{pub}}, v_{\text{exp}})$, strictly reducing recall $R_g$ while leaving internal nodes unchanged ($\Delta\text{F1}_g < 0$). Because $\text{Gate}_{G_5}(\mathcal{B}_{\text{local}})$ requires $\exists b \in \mathcal{B}_{\text{local}}$ with $\text{bnd}(b) = 0 \vee \text{supp}(b) \ge 2$, every query whose candidate set $\mathcal{B}_{\text{local}}$ contains only single-source open-boundary overrides is deterministically rejected ($\mathcal{R}_q \leftarrow \varnothing$), guaranteeing $\hat{T}_q(G_5) \equiv \hat{T}_q(\text{No-Ret})$ and $\Delta\text{F1}_g = 0$ across all unconstrained single-source boundary queries.

(3) *Terminal signal-delay separation under $G_5^{kin}$:* For $\text{supp}(b) \ge 2$ at $i_b = m$ with $\text{trunc}_m(b) = 1$, pass-through teacher intervals $e \in \mathcal{E}_{\text{pass}}(b)$ ($c^{\text{teacher}}_{\text{next}} \neq \varnothing$) traversed $v_{\text{prev}} \to v_{\text{exp}}$ ($v_{\text{prev}} = v_{c_{m-1},1}$) before continuing past the intersection. A query vehicle terminating at $c_m$ validly stops at the upstream queue node $v_{\text{exp}}$ (rather than exiting via $v_{\text{pub}}$) if and only if its observed interval duration $\Delta t_{m-1\to m} = t_m - t_{m-1}$ includes red-light waiting time exceeding the teachers' pass-through duration $\max_{e\in\mathcal{E}_{\text{pass}}(b)} \Delta t_{\text{teacher}}(e)$ and satisfies $d_{\text{SP}}(v_{\text{prev}}, v_{\text{exp}})/\max(\Delta t_{m-1\to m}, \epsilon) \le v_{\text{stop}} = 5.0\,\text{m/s}$. Both quantities are computed directly from pre-call inputs $(O_q, \mathcal{G}, \mathcal{E}_{\text{bank}})$ in $O(|\mathcal{B}_{\text{local}}|)$ time (< 1 ms) without ground-truth leakage or extra LLM calls. □

## B.2 Proof of Proposition 2

**Proof.** Fix a deterministic tie-breaking rule on equal-length shortest paths in $\mathcal{G}$ (lexicographic node-ID order, matching C2's bounded Dijkstra priority queue). By Definition 1 and Eq. (8) of the main paper, for every index pair $(a, b) \in \mathcal{I}_q = \{(i, i+1)\}_{i=1}^{m-1} \cup \{(i-1, i+1)\}_{i=2}^{m-1}$ and every candidate approach pair $(u, v) \in \Gamma_2(c_a) \times \Gamma_2(c_b)$, all vertices $V_{\text{SP}}(u, v)$ (alongside candidate via-nodes $\mathcal{W}_q$) belong

to $\mathcal{V}_q^{\text{core}} \subseteq \mathcal{N}_1^{\pm}(\mathcal{V}_q^{\text{core}}) = \mathcal{V}_q$, and all induced directed edges $\mathcal{E}_q = \mathcal{E} \cap (\mathcal{V}_q \times \mathcal{V}_q) \supseteq E_{\text{SP}}(u,v)$ along $P_{\text{SP}}^{\mathcal{G}}(u,v)$ are included in $\mathcal{G}_q = (\mathcal{V}_q, \mathcal{E}_q) \subseteq \mathcal{G}$. Because edge weights $\ell(e) > 0$ are identical on $\mathcal{E}_q \subseteq \mathcal{E}$ and no directed path in $\mathcal{G}_q \subseteq \mathcal{G}$ can have length strictly smaller than $d_{\text{SP}}^{\mathcal{G}}(u,v)$, deterministic tie-breaking yields $d_{\text{SP}}^{\mathcal{G}_q}(u,v) = d_{\text{SP}}^{\mathcal{G}}(u,v)$ and exact path equality $P_{\text{SP}}^{\mathcal{G}_q}(u,v) = P_{\text{SP}}^{\mathcal{G}}(u,v)$ for all $(u,v) \in \Gamma_2(c_a) \times \Gamma_2(c_b)$.

Next, consider C2's three-tier execution on $\mathcal{G}_q$ for any planned interval $(u \to \tilde{w}_{i,1} \to \tilde{w}_{i,2} \to v)$ with $(u,v) \in \Gamma_2(c_a) \times \Gamma_2(c_b)$: (i) Under Tier 1, if the proposed via-nodes lie in $\mathcal{V}_q$ and directed paths $P_{\text{SP}}^{\mathcal{G}_q}(u, \tilde{w}_{i,1})$, $P_{\text{SP}}^{\mathcal{G}_q}(\tilde{w}_{i,1}, \tilde{w}_{i,2})$, and $P_{\text{SP}}^{\mathcal{G}_q}(\tilde{w}_{i,2}, v)$ exist and satisfy C2's speed, span, and zero-detour bounds, every edge in the concatenated path belongs to $\mathcal{E}_q \subseteq \mathcal{E}$ (moreover, whenever $\tilde{w} \in V_{\text{SP}}(u,v)$, optimal substructure guarantees that concatenating $P_{\text{SP}}^{\mathcal{G}_q}(u,\tilde{w})$ and $P_{\text{SP}}^{\mathcal{G}_q}(\tilde{w},v)$ recovers $P_{\text{SP}}^{\mathcal{G}}(u,v)$). (ii) If any proposed via-node $\tilde{w}_{i,k}$ is outside $\mathcal{V}_q$, unreachable in $\mathcal{G}_q$, or violates kinematic/detour bounds, C2 deterministically drops $\tilde{w}_{i,k}$ and falls back to Tier 2's direct shortest path $P_{\text{SP}}^{\mathcal{G}_q}(u,v) = P_{\text{SP}}^{\mathcal{G}}(u,v) \subseteq \mathcal{E}_q$. (iii) If $(v_i^*, v_{i+1}^*)$ itself violates kinematic or mid-block zero-detour bounds, C2 trims isolated boundary spikes ($i \in \{1, m\}$) or invokes Tier 3 on interior $(i-1, i+1) \in \mathcal{I}_q$, expanding $P_{\text{SP}}^{\mathcal{G}_q}(v_{i-1}^*, v_{i+1}^*) = P_{\text{SP}}^{\mathcal{G}}(v_{i-1}^*, v_{i+1}^*) \subseteq \mathcal{E}_q$. Thus every emitted interval sequence $(w_{i,0}, \ldots, w_{i,L_i})$ is a valid directed path in $\mathcal{E}_q \subseteq \mathcal{E}$, and boundary anchor deduplication $(w_{i,L_i} = w_{i+1,0})$ preserves directed adjacency across interval joins, guaranteeing $(u_l, u_{l+1}) \in \mathcal{E}$ for all $1 \le l < |\hat{I}_q|$ and ensuring 100% road-network validity (0% segment violations). □

# C PROMPT CONTRACT AND ROBUSTNESS

## C.1 Prompt and JSON Schema Contracts

Each triggered cluster $O_q$ is serialized into three prompt blocks: `[CLUSTER_TRACK_FACTS]` (ordered observation table, Top-2 candidate nodes $\Gamma(c_i)$, output contract, and corridor subgraph $\mathcal{G}_q$), `[PAIR_COMPARISON_FACTS]` (shortest-path lengths, implied speeds, and candidate via-intersections), and (when accepted by $G_5/G_5^{\text{kin}}$) `[MAPPING_EXPERIENCE_JSON]` ($\le$ 4,000 chars). Under greedy decoding (temperature 0.0, seed 0), the LLM emits one JSON object adhering to the C2 road-intent schema. On held-out Test48 ($N = 48$, Table 3 of the main paper), decoupling intent planning from C2's bounded Dijkstra verifier cuts disconnected segments from 88.5% (108/122 under Direct-LLM and Prompt-Rules) to 0.0% (0/699 and 0/519), lifting Macro F1 from 0.3507 to **0.8540** (47W/0T/1L):

```
{
"camera_node_choices": [{"camera_id": 33, "road_node": 281}],
"tracks": [{"obs_ids": ["record:242216", "record:242217"],
"interval_intents": [{"from_obs_id": "record:242216",
"to_obs_id": "record:242217", "via_nodes": [6232]}]}],
"unassigned_obs_ids": []
}
```

## C.2 Tier Ablation and Multi-Seed Robustness

Tables 2 and 3 (complementing Figure 7b of the main paper) evaluate C2 verification across UrbanVehicle Val/Test ($N = 400$) and 5-seed false-positive camera outlier injection ($\rho_{\text{noise}} \in \{5\%, 10\%, 20\%\}$):

(1) **Incremental verification gains (Table 2):** Tier 1 recovers intermediate links missed by Nearest-Node anchors, lifting LCSS by +30.7 to +32.0 points and Macro F1 from 0.4133 → 0.6332 (Val) and 0.4050 → 0.6123 (Test), whereas Tier 2 ($d_{\max}$ = 1,200 m, $v_{\max}$ = 25 m/s) prunes infeasible jumps to reach 0.7498 (+11.66 pts) and 0.7238 (+11.15 pts). Integrating C1's gated transition retrieval and C3 routing (**Full C1–C3**) reaches **0.7686** (Val) and **0.7515** (Test), concentrating +9.18 and +13.88 F1 points on triggered multi-approach cohorts (Val37 and Test48) while bypassing clean single-corridor trips locally.

(2) **Widening Tier-3 margin under outlier noise (Table 3):** Across 5 seeds ($\rho_{\text{noise}}$ = 5% → 10% → 20%), rejecting outliers via Tier 2 alone fragments trajectories at corrupted interior points, dropping Tier 1+2 Macro F1 by −2.64, −4.79, and −9.12 points (0.6326 ± 0.0068 at 20%). Tier 3's two-hop bypass ($v_{i-1}^* \to v_{i+1}^*$) bridges excised spikes, widening Tier 1+2+3's lead over Tier 1+2 from +2.58 (5%) to +4.26 (10%) and +7.42 points (20%) while limiting degradation vs. clean to −0.26, −0.73, and −1.90 points (0.7068 ± 0.0025).

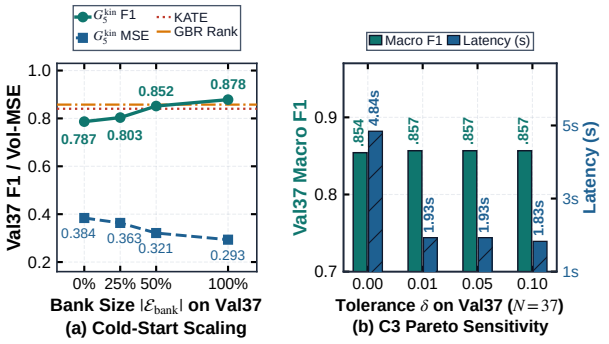



**Figure 2: Cold-start experience-bank scaling and out-of-sample C3 Pareto routing sensitivity: (a) monotonic zero-loss Macro F1 and Vol-MSE improvements under $G_5^{\text{kin}}$ on held-out Val37 ($N = 37$) as $|\mathcal{E}_{\text{bank}}|$ scales from 0% (cold-start C2 floor) to 100% (90 experiences), compared against KATE Cosine and GBR Ranker; (b) out-of-sample C3 quality–latency trade-off (Macro F1 on left axis; Mean Latency on right axis) across quality tolerance $\delta \in \{0.00, 0.01, 0.05, 0.10\}$ on held-out Val37 ($N = 37$, GBR routers trained on source-disjoint historical call logs).**

# D ROUTER FEATURES AND PRUNING

## D.1 Router Features and Cold-Start Scaling

Table 4 specifies the 18-dimensional pre-call feature vector $\mathbf{x}_q \in \mathbb{R}^{18}$ (9 transformed structural/kinematic attributes and 9 missingness flags) extracted in $O(m)$ time (< 1 ms) before dispatch. Figure 2a evaluates cold-start scaling on held-out Val37 ($N = 37$) as $|\mathcal{E}_{\text{bank}}|$ grows from 0% (zero-shot C2 floor) through 25% (22), 50% (45), and 100% (90 experiences): $G_5^{\text{kin}}$ raises Macro F1 monotonically (0.7866 → 0.8035 → 0.8517 → **0.8784**; Vol-MSE 0.3840 → **0.2935**) with zero losses (0L) at every fraction, whereas KATE and GBR rankers incur 8–15 regressions. Figure 2b confirms that relaxing C3's tolerance from $\delta = 0.00$ (4.84 s) to $\delta = 0.01$ (1.93 s) maintains Macro F1 (0.8567) and 1.83–1.93 s latency across $\delta \in [0.01, 0.10]$.

**Table 2: Incremental ablation of C2 corridor verification and full C1–C3 integration across UrbanVehicle Val ($N = 400$) and UrbanVehicle Test ($N = 400$) evaluated under the unified five-metric trajectory recovery suite.**

| Evaluation Split ($N$) | Verification Tier / Recovery Configuration | Macro F1 ↑ | Micro F1 ↑ | STLC ↑ | LCSS ↑ | Vol-MSE ↓ |
|---|---|---|---|---|---|---|
| UrbanVehicle Val ($N = 400$) | w/o C2 (Nearest-Node Anchors Only) | 0.4133 | 0.3878 | 0.6938 | 0.3155 | 0.6638 |
| | C2 Tier 1 Only (Unconstrained Shortest-Path) | 0.6332 | 0.6912 | 0.8297 | 0.6356 | 0.5037 |
| | C2 Tier 1+2 (+ Kinematic Span & Speed Gate) | 0.7498 | 0.7693 | 0.8844 | 0.7474 | 0.5003 |
| | **Full (C1–C3)** | **0.7686** | **0.7884** | **0.8908** | **0.7643** | **0.4919** |
| UrbanVehicle Test ($N = 400$) | w/o C2 (Nearest-Node Anchors Only) | 0.4050 | 0.3885 | 0.6872 | 0.3083 | 0.6614 |
| | C2 Tier 1 Only (Unconstrained Shortest-Path) | 0.6123 | 0.6676 | 0.8161 | 0.6155 | 0.5195 |
| | C2 Tier 1+2 (+ Kinematic Span & Speed Gate) | 0.7238 | 0.7473 | 0.8692 | 0.7230 | 0.5127 |
| | **Full (C1–C3)** | **0.7515** | **0.7749** | **0.8793** | **0.7490** | **0.5010** |

**Table 3: Multi-seed robustness (mean ± std across 5 independent random seeds) under synthetic false-positive camera outlier contamination ($\rho_{\text{noise}} \in \{5\%, 10\%, 20\%\}$) on UrbanVehicle Test ($N = 400$), alongside Macro F1 change (ΔMacro F1) relative to each variant's unperturbed baseline.**

| Outlier Contamination Rate | C2 Verification Variant | Macro F1 ↑ | Micro F1 ↑ | STLC ↑ | LCSS ↑ | ΔMacro F1 vs. Clean |
|---|---|---|---|---|---|---|
| $\rho_{\text{noise}} = 5\%$ Contamination (UV Test $N = 400$, 5 Seeds) | C2 Tier 1 Only (Unconstrained Shortest-Path) | 0.5982 ± 0.0033 | 0.6575 ± 0.0028 | 0.8078 ± 0.0017 | 0.5953 ± 0.0035 | −1.41 pts |
| | C2 Tier 1+2 (w/o Tier-3 2-Hop Spike Bypass) | 0.6974 ± 0.0031 | 0.7239 ± 0.0026 | 0.8531 ± 0.0021 | 0.6899 ± 0.0036 | −2.64 pts |
| | **C2 Tier 1+2+3 (+ Two-Hop Spike Bypass)** | **0.7232 ± 0.0008** | **0.7475 ± 0.0006** | **0.8685 ± 0.0005** | **0.7220 ± 0.0008** | **−0.26 pts** |
| $\rho_{\text{noise}} = 10\%$ Contamination (UV Test $N = 400$, 5 Seeds) | C2 Tier 1 Only (Unconstrained Shortest-Path) | 0.5922 ± 0.0071 | 0.6529 ± 0.0029 | 0.8011 ± 0.0032 | 0.5832 ± 0.0067 | −2.01 pts |
| | C2 Tier 1+2 (w/o Tier-3 2-Hop Spike Bypass) | 0.6759 ± 0.0042 | 0.7049 ± 0.0025 | 0.8384 ± 0.0021 | 0.6618 ± 0.0053 | −4.79 pts |
| | **C2 Tier 1+2+3 (+ Two-Hop Spike Bypass)** | **0.7185 ± 0.0017** | **0.7429 ± 0.0014** | **0.8652 ± 0.0008** | **0.7159 ± 0.0019** | **−0.73 pts** |
| $\rho_{\text{noise}} = 20\%$ Contamination (UV Test $N = 400$, 5 Seeds) | C2 Tier 1 Only (Unconstrained Shortest-Path) | 0.5681 ± 0.0080 | 0.6320 ± 0.0038 | 0.7856 ± 0.0055 | 0.5502 ± 0.0098 | −4.42 pts |
| | C2 Tier 1+2 (w/o Tier-3 2-Hop Spike Bypass) | 0.6326 ± 0.0068 | 0.6650 ± 0.0058 | 0.8116 ± 0.0046 | 0.6083 ± 0.0091 | −9.12 pts |
| | **C2 Tier 1+2+3 (+ Two-Hop Spike Bypass)** | **0.7068 ± 0.0025** | **0.7309 ± 0.0030** | **0.8570 ± 0.0018** | **0.7014 ± 0.0033** | **−1.90 pts** |

**Table 4: Pre-call feature vector $\mathbf{x}_q \in \mathbb{R}^{18}$ (9 transformed structural/kinematic attributes +9 missingness flags) for the C3 router (Section 3.3.1 of the main paper).**

| Idx | Pre-Call Feature Attribute | Pre-Call Transform & Operational Definition |
|---|---|---|
| 1, 10 | `traj_len` (+ missing flag) | $\log(1 + \max(x, 0))$ of cluster observation count $m$ |
| 2, 11 | `time_span` (+ missing flag) | $\log(1 + \max(x, 0))$ of cluster duration $t_m - t_1$ (s) |
| 3, 12 | `spatial_range` (+ missing flag) | $\log(1 + \max(x, 0))$ of camera coordinate bounding span |
| 4, 13 | `branching_score` (+ missing flag) | $\text{clip}(x, 0, 1)$ road-graph candidate branching ratio |
| 5, 14 | `missing_ratio` (+ missing flag) | $\text{clip}(x, 0, 1)$ estimated unobserved intermediate hop ratio |
| 6, 15 | `jump_distance` (+ missing flag) | $\log(1 + \max(x, 0))$ of mean inter-camera jump distance |
| 7, 16 | `input_char_length` (+ missing flag) | $\log(1 + \max(x, 0))$ of serialized prompt character count |
| 8, 17 | `retrieval_count` (+ missing flag) | $\log(1 + \max(x, 0))$ of retrieved transition candidates |
| 9, 18 | `retrieval_similarity` (+ missing flag) | $\text{clip}(x, 0, 1)$ top retrieved structural similarity score |

## D.2 Subgraph Pruning and Local CPU Profile

Table 5 audits corridor subgraph compression ($\mathcal{G} \to \mathcal{G}_q$, Eq. (8) of the main paper) across two-anchor prompts (34.9 nodes, 21.9 KB) and full UrbanVehicle Test clusters ($N = 400$, 99.6 nodes), alongside local CPU overhead. Pruning $\mathcal{G}$ (8,270 nodes, 9,566 edges) to $\mathcal{G}_q$ removes 98.8%–99.6% of nodes, 94.0% of prompt bytes, and 96.3%–97.4% of input tokens with 100.0% (547/547) shortest-path parity (cutting unrouted Qwen3.5-Plus latency from 13.53 s to 6.60 s and routed C3 latency to 1.98 s). Extending $\mathcal{I}_q$ from consecutive pairs $\{(i, i+1)\}$ (88.0 nodes) to two-hop pairs $\{(i-1, i+1)\}$ (99.6 nodes) preserves linear $O(m)$ search while adding only 0.06 ms (0.65 → 0.71 ms) to retain all Tier-3 bypass paths without $O(m^2)$ all-pairs search (113.6 nodes, 0.80 ms).

Across Stages 1–3, local non-LLM CPU execution totals 1.81 ms per cluster (1.08 ms Stage 1 gating, 0.71 ms Stage 2 $\mathcal{G}_q$ pruning, and 0.016 ms Stage 3 C2 Dijkstra verification), or $< 0.10\%$ of the 1.98 s routed latency on Test48.

**Table 5: Corridor subgraph pruning ($\mathcal{G} \to \mathcal{G}_q$, Eq. (8) of the main paper) graph compression across two-anchor interval prompts (164 cases, 547 intervals) and full multi-observation clusters ($N = 400$), shortest-path equivalence, and stage-wise non-LLM local CPU execution profile on UrbanVehicle Test.**

| Diagnostic Audit Scope | Graph Scope / Local Pipeline Stage | Mean Active Nodes | Mean Active Edges | Edge Reduction | Shortest-Path Parity | Execution Time (p50) |
|---|---|---|---|---|---|---|
| Two-Anchor Corridor Prompt (164 Cases, 547 Intervals) | Unpruned Full City Graph $\mathcal{G}$ (361.4 KB prompt) | 8,270.0 (100.0%) | 9,566.0 (100.0%) | 0.00% (Ref.) | 100.0% (547/547 Ref.) | 13.53 s (Test48, $\mathcal{G}$) |
| | **Pruned Corridor Subgraph $\mathcal{G}_q$ (21.9 KB prompt)** | **34.9 (0.42%)** | **39.1 (0.41%)** | **−99.59%** | **100.0% (547/547 Exact)** | **6.60 s ($\mathcal{G}_q$) / 1.98 s (Routed)** |
| Cluster-Wide $\mathcal{G}_q$ Pair Scope (UV Test $N = 400$ Multi-Obs.) | Consecutive Pairs Only ($O(m)$): $\mathcal{I}_q = \{(i, i+1)\}$ | 88.0 (1.06%) | 98.3 (1.03%) | −98.97% | Tier 1–2 Only (Omits 2-Hop) | 0.65 ms (0.25 ms) |
| | **Consecutive + 2-Hop ($O(m)$): $\mathcal{I}_q$ (Default Eq. (8))** | **99.6 (1.20%)** | **112.4 (1.18%)** | **−98.82%** | **100.0% (Full Tier 1–3)** | **0.71 ms (0.26 ms)** |
| | All Forward Pairs ($O(m^2)$): $\mathcal{I}_q = \{(a, b) : a < b\}$ | 113.6 (1.37%) | 129.4 (1.35%) | −98.65% | 100.0% (Full Tier 1–3) | 0.80 ms (0.25 ms) |
| Local Non-LLM CPU Profile (UV Test $N = 400$ Clusters) | Stage 1 Kinematic & $G_5^{\text{kin}}$ Gate ($\gamma_{\text{inv}}$) | — | — | — | Exact Pre-Call Filter | 1.08 ms (0.26 ms) |
| | Stage 2 Corridor Subgraph Pruner ($\mathcal{G} \to \mathcal{G}_q$) | 99.6 (1.20%) | 112.4 (1.18%) | −98.82% | 100.0% Exact SP Parity | 0.71 ms (0.26 ms) |
| | Stage 3 Bounded C2 Dijkstra Verifier | 99.6 (1.20%) | 112.4 (1.18%) | −98.82% | 100.0% Valid Edges | 0.016 ms (0.010 ms) |
| | **Total Non-LLM Local CPU (Stages 1–3)** | — | — | — | **< 0.10% of Routed Latency** | **1.81 ms (0.53 ms)** |